

CCS3700 Embedded Systems Programming

Introduction

Dr. Khairul Azhar Kasmiran

Department of Computer Science
Faculty of Computer Science and Information
Technology

Room C3-14 (temporarily C1-24), Block C

▶ Email: k.azhar@upm.edu.my
WhatsApp: 012-2948146

Lecture Outlines

- ▶ Basic concepts of embedded systems
- ▶ Conventional system vs. embedded systems
- ▶ Examples of embedded systems
- ▶ Microprocessor and microcontroller

Learning Outcome

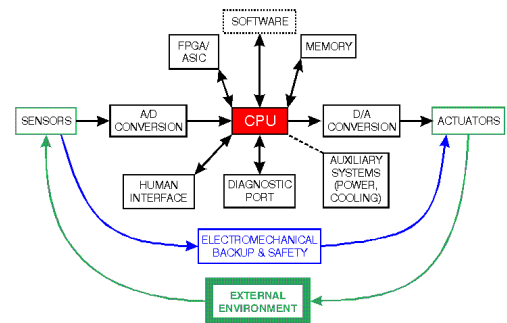
Explain the concepts of embedded systems (C3)

Discuss the usefulness of embedded systems in particular application areas (A2).

State the characteristics of embedded operating system (P2)

Embedded Systems

An **embedded system** is a device or module designed for a specific purpose and based on some computational (software modifiable) element. One of the key features of an embedded system is **real-time** operation.



Embedded System

- ▶ A combination of hardware and software designed to perform a dedicated function
- ▶ Often, embedded systems are part of a larger system or product,
 - e.g., antilock braking system in a car
- ▶ Embedded systems are tightly coupled to their environment → **imposes real-time constraints** by the need to interact with the environment

Embedded Systems



- **Computers used as part of a larger system**
 - That usually doesn't look like a computer
 - That usually controls physical devices
- **Often reliability is critical**
 - "Critical" as in "if the system fails someone might die"
- **Often resources (memory, processor capacity) are limited**
- **Often real-time response is essential**

The landscape of embedded systems

- ▶ **Lots of embedded applications require very high performance:**
 - Communications.
 - Multimedia.
 - Graphics.
- ▶ **Must also meet strict design goals:**
 - Real-time performance.
 - Power/energy consumption.
 - Cost.
- ▶ **How does the embedded system design constraints differ from general purpose computer system design constraints?**

Overview

- ▶ **Embedded systems**
 - Computing systems embedded within electronic devices
 - Hard to define. Nearly any computing system other than a desktop computer
 - Billions of units produced yearly, versus millions of desktop units
 - Perhaps 50 per household and per automobile



Example of embedded systems applications

Anti-lock brakes
Auto-focus cameras
Automatic teller machines
Automatic toll systems
Automatic transmission
Avionic systems
Battery chargers
Camcorders
Cell phones
Cell-phone base stations
Cordless phones
Cruise control
Curbside check-in systems
Digital cameras
Disk drives
Electronic card readers
Electronic instruments
Electronic toys/games
Factory control
Fax machines
Fingerprint identifiers
Home security systems
Life-support systems
Medical testing systems

Modems
MPEG decoders
Network cards
Network switches/routers
On-board navigation
Pagers
Photocopiers
Point-of-sale systems
Portable video games
Printers
Satellite phones
Scanners
Smart ovens/dishwashers
Speech recognizers
Stereo systems
Teleconferencing systems
Televisions
Temperature controllers
Theft tracking systems
TV set-top boxes
VCR's, DVD players
Video game consoles
Video phones
Washers and dryers



Some common characteristics of embedded systems

- ▶ **Single-functioned**
 - Executes a single program, repeatedly
- ▶ **Tightly-constrained**
 - Low cost, low power, small, fast, etc.
- ▶ **Reactive and real-time**
 - Continually reacts to changes in the system's environment
 - Must compute certain results in real-time without delay

10

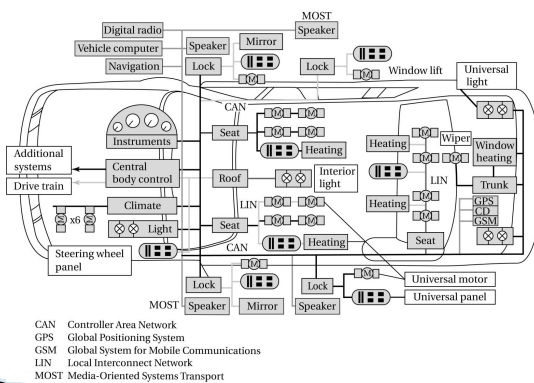
Sample Application Domains

- ▶ **Wireless communication systems**
 - Radios and networking
 - Tower control (Airport)
- ▶ **Multimedia**
 - Image and video processing
- ▶ **Real-time control**
 - Automotive and avionic systems
 - Automated Teller Machine (ATM)

Networks and embedded systems

- ▶ An increasing number of embedded systems connect to the Internet.
 - Resource management.
 - Security.
 - **Called Internet of Things (IoT) – Example?**
- ▶ Many specialized networks have been developed for embedded systems:
 - Automotive.
 - Device control.

Automobiles as distributed embedded systems





Automotive and aviation electronics

- ▶ Some functions are safety-critical.
- ▶ Must operate in real-time.
- ▶ Must fit within power budget (limited by generator).
- ▶ Must be lightweight to fit within vehicle weight budget.
- ▶ How would processors for these types of systems differ from those for multimedia systems?

Automotive electronics/avionics uses

- ▶ Operator vs. passenger: Passenger operations are less critical, more varied (TV, Internet, etc.).
- ▶ Control vs. instrumentation: Instruments report on the vehicle, control closes the loop, diagnostic

Sensor networks

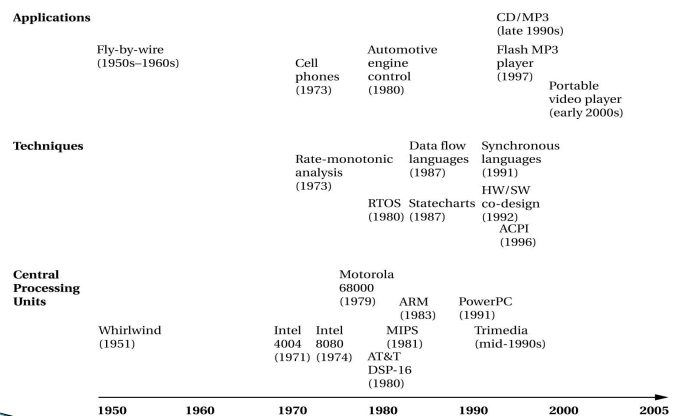
- ▶ Used to gather, process data in the field.
- ▶ Ad-hoc networks: must set themselves up without intervention of network manager.
- ▶ Often battery powered, very tight energy budget.
- ▶ Generally wirelessly networked.

Embedded Systems

- ▶ **Hard real time**
 - Response must occur before the deadline
- ▶ **Soft real time**
 - Response should occur before the deadline most of the time
- ▶ Often there are plenty of resources to handle the common cases
 - But crises happen and must be handled
- ▶ **Predictability is key**
- ▶ **Correctness** is even more important than usual
 - "correctness" is not an abstract concept
 - "but I assumed that the hardware worked correctly" is no excuse
 - Over a long time and over a large range of conditions, it simply doesn't

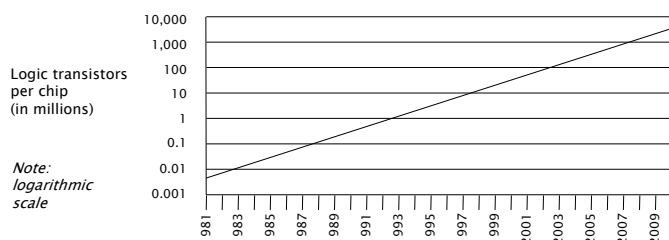
"A real time system is one in which the correctness of the computations not only depends upon the logical correctness of the computation but also upon the time at which the result is produced. If the timing constraints of the system are not met, system failure is said to have occurred."

History of embedded systems

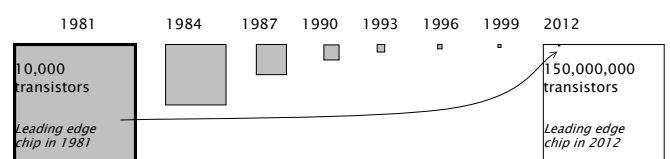


Moore's law

- ▶ The most important trend in embedded systems
 - Predicted in 1965 by Intel co-founder Gordon Moore
- ▶ **IC transistor capacity has doubled roughly every 18 months for the past several decades**



Graphical illustration of Moore's law



- ▶ Something that doubles frequently grows more quickly than most people realize!
 - A 2012 chip can hold about 15,000 1981 chips inside itself

Disciplines in embedded system

Core areas:

- Real-time computing
- Hardware/software co-design.

Closely related areas:

- Computer architecture.
- Software engineering.
- Low-power design.
- Operating systems.
- Networking.
- Secure and reliable computing.
- Programming languages and compilers.

The constraints of embedded system are also the challenges for the system design and development.

- Resource constraint:** Embedded systems are constrained for their size, power capacity (may be battery operated), limited memory capacity (especially RAM size), CPU speed and function capacity
- Time constraints:** Real-time system has response time deadline to meet.
- Environment constraint:** extreme operation condition such as high or low temperature, high pressure, high humidity, under water, under collision, etc.
- Cost Constraint:** Cost reduction of products has huge impact on the market competition because the large volume of products For high volume systems such as MP3 players or PDAs, minimizing cost is usually the primary design consideration. The selected hardware should be just "good enough" to cover the necessary functions.
- Time to market constraint:** Design and development cycle is very limit in order to beat the competent in the market.

What is a Microcontroller

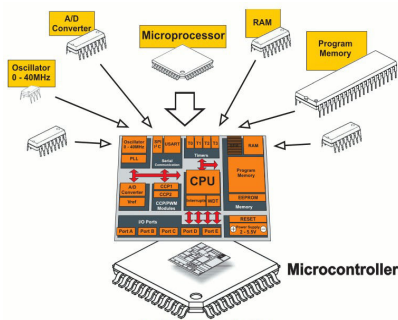
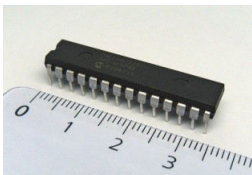


Fig. 1-1 Microcontroller versus Microprocessor

- A small computer on a single chip
 - containing a processor, memory, and input/output
- Typically **"embedded"** inside some device that they control
- A microcontroller is often small and low cost

Some Popular Microcontrollers...

- Intel 8051
- Microchip Technology PIC
- Atmel AVR / Arduino**
- Texas Instruments MSP430 (16-bit)



Characteristics of Embedded System

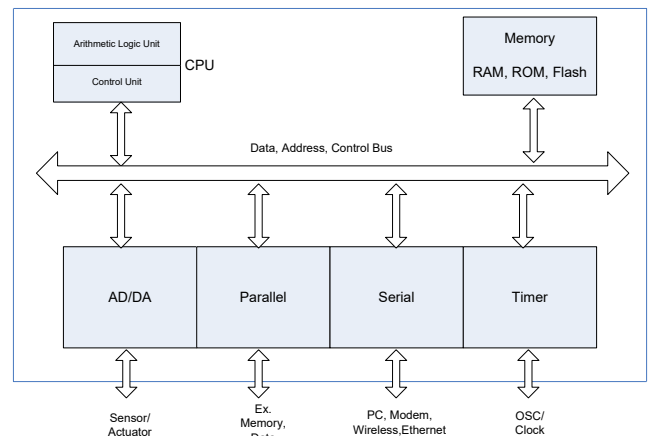
- Reliability** is a very important character for embedded systems. Many systems work in mission critical and life-threatening environment where system reliability is crucial.
- Embedded systems are often embedded in machines that are expected to **run continuously** for years without faults, and tolerate the errors by themselves if an error occurs.
- Therefore the **software is usually developed and tested more thoroughly** than that for personal computers.
- Most embedded systems cannot be shut down for maintenance, or it is very difficult to repair such as aerospace systems. Some system will lose large amounts of money when shut down: Telephone switches, factory controls, bridge and elevator controls, funds transfer and market making, automated sales and service.

Architecture

Central Processing Unit (CPU) is the heart of microprocessor(μP) or microcontroller(μC).

A **microprocessor** is a **stand-alone CPU chip** and memory and I/O ports can be custom designed and expanded. A microprocessor chip is a single integrated circuit intended to operate for general purpose and can be embedded into embedded electronics hardware.

A **microcontroller** is an **integrated chip** which comes with built-in memory, I/O ports, timers, and other components. Most embedded systems are built on microcontrollers which run faster than custom built system with microprocessor because all components are integrated within a single chip. There is a wide variety of microprocessors and microcontroller available.



Microprocessor vs. Microcontroller

Microprocessor

- CPU is stand-alone, RAM, ROM, I/O, timer are separate
- designer can decide on the amount of ROM, RAM and I/O ports.
- expensive
- versatility
- general-purpose
- High processing power
- High power consumption
- Instruction sets focus on processing-intensive operations
- Typically 32/64 – bit
- Typically deep pipeline (5-20 stages)

Microcontroller

- CPU, RAM, ROM, I/O and timer are all on a single chip
- fixed amount of on-chip ROM, RAM, I/O ports
- for applications in which cost, power and space are critical
- single-purpose (control-oriented)
- Low processing power
- Low power consumption
- Bit-level operations
- Instruction sets focus on control and bit-level operations
- Typically 8/16 bit
- Typically single-cycle/two-stage pipeline

Small and Many!

- ▶ Most embedded systems use 4-, 8-, or 16-bit processors. 32-bit used for intensive applications like printer controls.
- ▶ 8-bit processors have about 64K of memory, that limits amount of code.
- ▶ “By 1990 a total of about 45 million recognizable computers (i.e., PCs, Macintosh, even CP/M systems) were in place. Yet over 1 billion microprocessors and microcontrollers were shipped in that year alone!”

Source: Ganssle, J. *The Art of Programming Embedded Systems*. Academic Press, 1992, San Diego

33

Single Board Computers



- **Little Board** (5.75 x 8.0 in.) -- complete systems on a single compact board, expandable with plug-on function modules



- **ISA "slot boards"** (full-length, 13.8 x 4.8 in.; half-length, 7.1 x 4.8 in.) -- IBM PC plug-in cards which could function as standalone SBCs backplanes)

- **PC/104 modules** (3.6 x 3.8 in.) -- compact, rugged, self-stacking modules featuring a reliable pin-and-socket board-to-board expansion bus

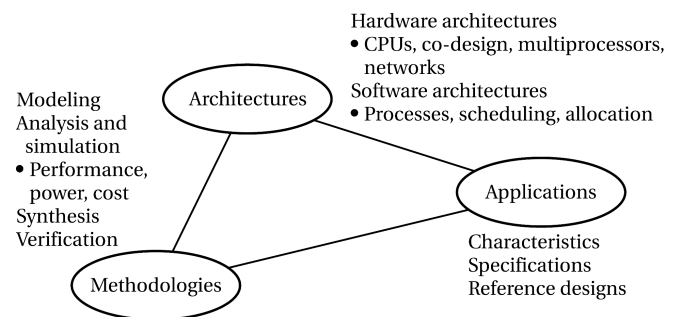
- **PC/104-Plus** -- PCI added to PC/104

- **EBX** -- PC/104-Plus added to Little Board

Designing embedded systems

- ▶ **No one architecture (hardware or software) can meet the needs of all applications.**
- ▶ **We need to be able to design a system for our target application or application domain:**
 - **Quickly and efficiently.**
 - **With reliable results.**

Aspects of embedded system design



Architectures

- ▶ Both hardware and software architectures are important.
- ▶ The structure of the system determines cost, power, and performance.
- ▶ Different application requirements lead us to different architectures.
 - ▶ You can't design the best embedded systems if you don't know anything about your application.
 - ▶ You can't be an expert in everything.
 - But a little knowledge goes a long way.
 - ▶ Domain expertise helps you make trade-offs:
 - Can the requirements be relaxed?
 - Can one requirement be traded for another?
 - How can the system be optimized?

How are embedded systems different than traditional software?

- ▶ Responding to sensors (Was this button pushed?)
- ▶ Turning on actuators (Turn on power to the boiler.)
- ▶ Real-time (Respond to temperature change within 3 seconds.)

Differences between embedded systems and conventional system

- ▶ Not dealing with only sequential code.
- ▶ Routine can stop at completion or in response to an external event.
- ▶ Many parts of system might be running concurrently.
- ▶ Safety-critical component of many systems.

Increased Complexity

- ▶ Systems are becoming more and more complex.
- ▶ Harder to think about total design.
- ▶ Harder to fix 'bugs.'
- ▶ Harder to maintain systems over time.
- ▶ Therefore, the traditional development process has to change,

Less Time to Develop

- ▶ In embedded systems, the total design cycle must decrease.
- ▶ Historically, design for automotive embedded systems takes 3-5 years to develop.
- ▶ Must become a 1-3 year development cycle.
- ▶ Must still be reliable and safe.

What an embedded system is NOT?

- ▶ Not a computer system that is used primarily for processing.
- ▶ Not a software system on a PC or Unix box.
- ▶ Not a traditional business or scientific application.



How are embedded systems different than traditional software?

- ▶ Responding to sensors (Was this button pushed?)
- ▶ Turning on actuators (Turn on power to the boiler.)
- ▶ Real-time (Respond to temperature change within 3 seconds.)

Differences between embedded systems and conventional system

- ▶ Not dealing with only sequential code.
- ▶ Routine can stop at completion or in response to an external event.
- ▶ Many parts of system might be running concurrently.
- ▶ Safety-critical component of many systems.

Differences from typical computer

- ▶ A variety of interfaces
 - enable the system to measure, manipulate, and interact with the external environment
 - human interface may be as simple as a flashing light or as complicated as real-time robotic vision
- ▶ Use of a diagnostic port for diagnosing the system
- ▶ Use of special purpose hardware to increase performance or safety
- ▶ Fixed-function or application-specific software



Basic Concepts of Embedded systems - Charge

- A property of particles that experience electromagnetic force
- Two kinds of charge: positive and negative
- Force due to charge obeys an inverse square law
- Charge is measured in coulombs
- Electrons and protons each have the same size charge (but of opposite polarity)
- Charge magnitude = 1.6×10^{-19} coulombs

Basic Concepts of Embedded systems - Current

- Current is charge in motion
- Most of the time we think about electrons moving through metallic wires
- The flow rate of charge is measured in coulombs/second or Amperes (Amps)
 - $\text{charge/time} = \text{coulombs/sec} = \text{Amperes}$
- $1 \text{ Amp} = 1/1.6 \times 10^{19} \text{ electrons / sec}$

Basic Concepts of Embedded systems - Voltage

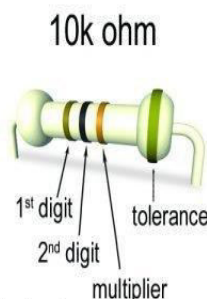
- **Voltage** is the driving force behind current
- Voltage is the *electrical potential energy* a charge has due to its position in space
 - potential energy per unit of charge
 - "path independent"
- Voltage is measured in Joules/Coulomb or **Volts (V)**
- Voltage is defined such that negatively charged particles are pulled towards higher voltages
- Potential energy can be converted into other forms of energy

Basic Concepts of Embedded systems - Resistance

- Resistance is a property of materials
- Resistors convert voltage to heat
- Ohm's law describes the relationship between voltage and current flow through a resistor
 - $V = I R$
 - V is the voltage across the resistor
 - I is the current flowing through the resistor
 - R is the resistance (depends upon the material)
- Resistance is measured in Ohms, Ω

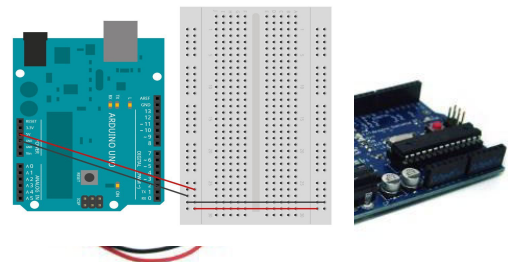
Basic Concepts of Embedded systems - Resistor Code

Color	Color Name	1 st digit 1 st stripe	2 nd digit 2 nd stripe	Multiplier 3 rd stripe	Tolerance 4 th stripe
	Black	0	0	x1	-
	Brown	1	1	x10	1%
	Red	2	2	x100	2%
	Orange	3	3	x1,000	3%
	Yellow	4	4	x10,000	4%
	Green	5	5	x100,000	-
	Blue	6	6	x1,000,000	-
	Violet	7	7	-	-
	Grey	8	8	-	-
	White	9	9	-	-
	Gold	-	-	x0.1	5%
	Silver	-	-	x0.01	10%



Copyright © 2006 All Spectrum Electronics

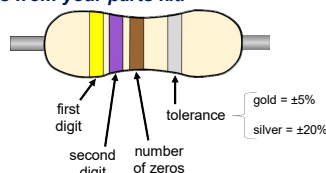
Basic Concepts of Embedded systems - Power Source



Basic Concepts of Embedded systems - Select Resistors

Find the 2200 and the 4700 resistors from your parts kit.

color	digit
black	0
brown	1
red	2
orange	3
yellow	4
green	5
blue	6
violet	7
gray	8
white	9



Example: 4700 resistor

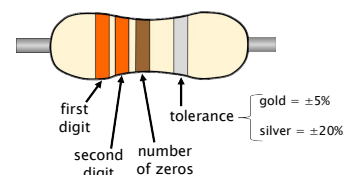
4 = yellow
7 = violet
Add 1 zero to 47 to make 470, so 1 = brown
So, 470 = **yellow, violet, brown**

Now, find the 2200 resistor colors.

Select Resistors

Find the 3300 and the 10k0 resistors from your parts kit.

color	digit
black	0
brown	1
red	2
orange	3
yellow	4
green	5
blue	6
violet	7
gray	8
white	9



Example: 3300 resistor:

3 = orange
3 = orange
Add 1 zero to 33 to make 330, so 1 = brown
So, 330 = **orange, orange, brown**

Now, find the 10k0 resistor.

Ohm's Law

Ohm's Law describes the direct relationship between the Voltage, Current and Resistance of a circuit.

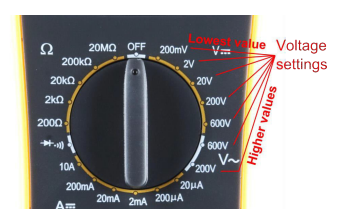
The three different forms of Ohm's Law are as follows:

$$V = I * R \quad I = V / R \quad R = V / I$$

Where V is Voltage, I is Current and R is Resistance.

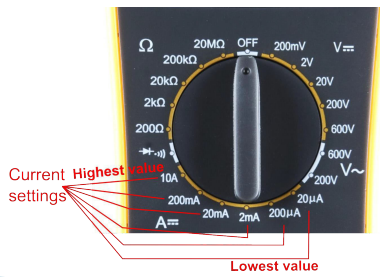
Ohm's Law

Voltage allows electricity to travel through all the components of the circuit. Voltage of your power source must be more than the total voltage drop of your circuit so the electricity can travel from power to ground.



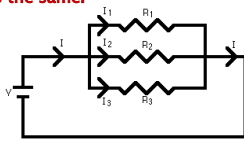
Ohm's Law

Current is the aspect of electricity that performs physical work or movement.



Parallel Circuit

A parallel circuit is a circuit in which the resistors are arranged with their heads connected together, and their tails connected together. The current in a parallel circuit breaks up, with some flowing along each parallel branch and re-combining when the branches meet again. **The voltage across each resistor in parallel is the same.**



A parallel circuit is shown in the diagram above. In this case the current supplied by the battery splits up, and the amount going through each resistor depends on the resistance. If the values of the three resistors are: **$R_1 = 8 \Omega$, $R_2 = 8 \Omega$ and $R_3 = 4 \Omega$** . With a 10 V battery, by $V = IR$ the total current in the circuit is: $I = V / R = 10 / 2 = 5 \text{ A}$. The individual currents can also be found using $I = V / R$. The voltage across each resistor is 10 V, so:
 $I_1 = 10 / 8 = 1.25 \text{ A}$
 $I_2 = 10 / 8 = 1.25 \text{ A}$
 $I_3 = 10 / 4 = 2.5 \text{ A}$

Kirchoff's Current Law (KCL)

The principle of conservation of electric charge implies that:

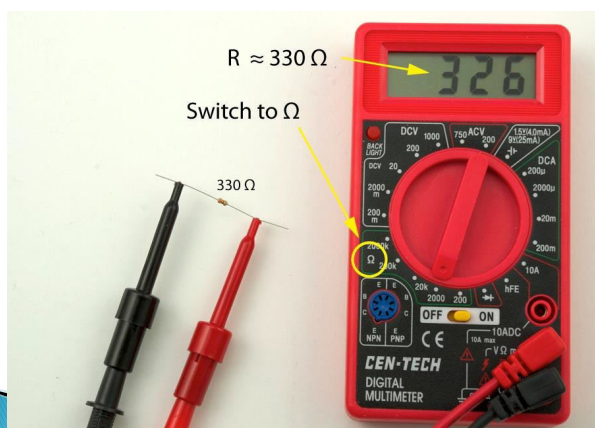
At any node (junction) in an electrical circuit, the sum of currents flowing into that node is equal to the sum of currents flowing out of that node, or: The algebraic sum of currents in a network of conductors meeting at a point is zero

Kirchoff's Current Law (KCL)

The algebraic sum of currents entering a node is zero.

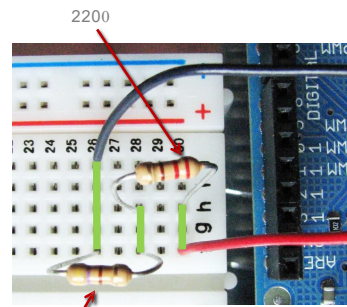
Current In = Current Out

Check Resistance of Resistors

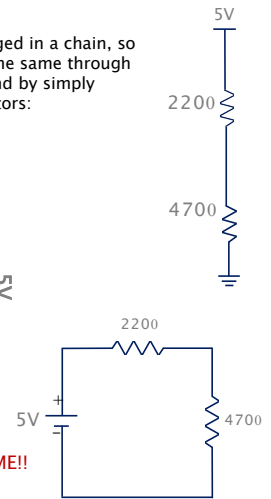


Series Circuit

A series circuit is a circuit in which resistors are arranged in a chain, so the current has only one path to take. The current is the same through each resistor. The total resistance of the circuit is found by simply adding up the resistance values of the individual resistors:



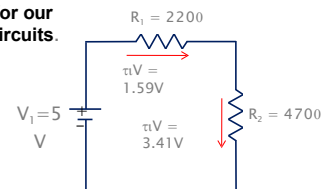
All of these circuits are the SAME!!



Kirchoff's Voltage Law (KVL)

Kirchoff's Voltage Law says that the algebraic sum of voltages around any closed loop in a circuit is zero – we see that this is true for our circuit. It is also true for very complex circuits.

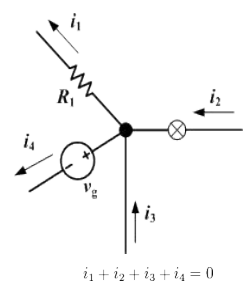
$$5V - 1.59V - 3.41V = 0$$



Notice that the 5V is DIVIDED between the two resistors, with the larger voltage drop occurring across the larger resistor.

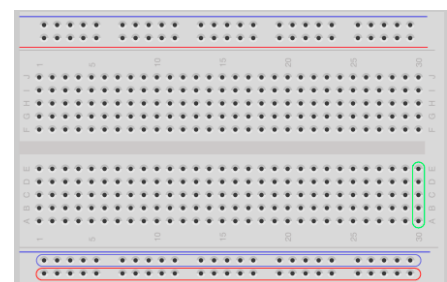
Kirchoff's Current Law (KCL)

We observe four currents "entering" the junction depicted as the bold black dot in Figure shown. Of course, two currents are actually exiting the junction, but for the purposes of circuit analysis it is generally less restrictive to consider what are in actuality positive currents flowing out of a junction to be negative currents flowing into that junction (mathematically the same thing). Doing so allows us to write Kirchhoff's law for this example as:



$$\sum_n i_n = i_1 + i_2 + i_3 + i_4 = 0$$

$$i_1 + i_2 + i_3 + i_4 = 0$$



Basic Concepts of Embedded systems – Breadboard

Breadboards are the way to rapid prototyping and this is the way to go here. Basically, breadboards consists of a piece of plastic with many perforations in which there are small pieces of conductors allowing to connect wires and components' leads inside.

The distance between two perforations is 2.54 mm (equal to 0.1") that is a standard;

for instance, dual in-line package integrated circuits' leads are all separated by this particular distance and thus, you can even put IC on breadboards.

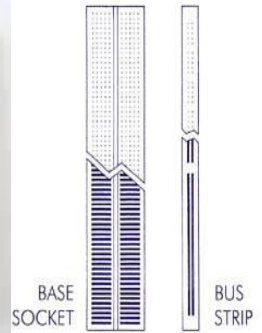
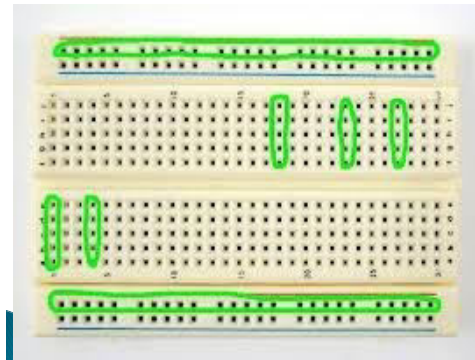
As we saw on the previous screenshot, there are buses and terminals strips.

Buses are series of five perforations in the central part and put in column for which the underlying conductors are connected. I have surrounded one bus with a green stroke.

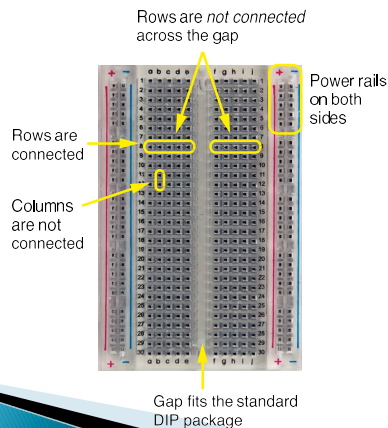
Terminals are special buses usually used for power supplying the circuit and appear in between blue and red lines. Usually, we use blue for ground lines and red for voltage source (5 V or 3.3 V in some cases). A whole line of terminals has its perforations all connected, providing voltage source and ground easily available on all the breadboard without having to use a lot of connection to the Arduino.

I surrounded 2 of the 4 terminals with red and blue strokes.

Breadboards provide one of the easiest ways of prototyping without soldering. It also means you can use and reuse your breadboards throughout the years!

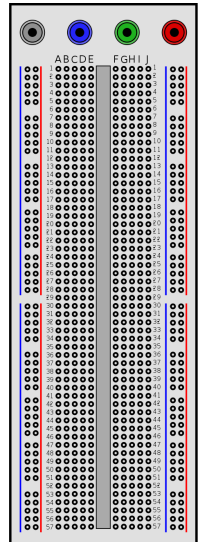


Building a circuit on a breadboard



Breadboard

- Each numbered row is connected internally, up to the middle separator.
- Each marked Bus Column is connected internally
 - Sometimes split in the middle
- **DIP ICs** straddle the middle
- Good for prototyping, bad for reliability and sensitive signals.



Breadboard Best Practices

- Neatness Counts. A
- lot. Color Code wires
- Use wires of approximately the correct length
- Start with ICs – always orient them the same way
 - Then add power and ground
 - Then add internal connections
 - Then add chip-to-chip connections
 - Then everything else

Following the same analogy, **resistors** are small components that slow down the flow of current. They are more resistive than any piece of wire you can use; they generally dissipate it as heat. They are two passive terminal components and aren't polarized, which means you can wire them in both directions.

Resistors are defined by their *electrical resistance* expressed in Ohms (the symbol is Ω). There is a direct mathematical relation between voltage measured at the resistor sides, current, and resistance known as the Ohm's law: $R = V / I$ where R the electrical resistance in Ohms, V the voltage in Volts, and I the current in Amperes.

For a constant value of voltage, if the resistance is high, the current is low and vice-versa. It is important to have that in mind.

On each resistor, there is a color code showing the resistance value. There are many types of resistors. Some have a constant resistance, some others can provide different resistance values depending on physical parameters such as temperature, or light intensity for instance.

A **potentiometer** is a variable resistor. You move a slider or rotate a knob and the resistance changes. I guess you begin to understand my point...

A **capacitor** (or **condenser**) is another type of component used very often. The direct analogy is the rubber membrane put in the pipe: no water can pass through it, but water can move by stretching it. They are also passive two-terminal components but can be polarized. Usually, small capacitors aren't.

We usually are saying that capacitors store potential energy by charging. Indeed, the rubber membrane itself stores energy while you stretch it; try to release the stretched membrane, it will find its first position.

Capacitance is the value defining each capacitor. It is expressed in Farads (the symbol is F).

